

PowerTrain Optimizer

Data Centre Power Economics Report

Captive & Hybrid procurement · Tariff / PPA stacks · CFE & carbon

Project	Default Data Centre - 250 MW
Study period	Jan–Dec 2026
Model version	2.0.0
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Report scope	Captive + Hybrid (equal analysis) · DISCOM = savings baseline only

This report summarises Year-1 power cost, bill, savings vs 100% DISCOM, NPV of savings, Architecture mix %, Storage tariff, and Hourly CFE Pass/Fail — matching the in-app analysis.

1. Project setup snapshot

Parameter	Value
IT load / PUE / Peak	100.0 MW · PUE 1.25 · Peak 125.0 MW
Load factor (S1 headline)	70%
Storage tariff (BESS)	■0.0000/kWh
Cost blend basis	TARGET POWER PCT
Annual RE target	90%
Hourly CFE target (24×7)	90%
Pass rule	Every hour's CFE% ≥ Hourly CFE target (min hourly CFE ≥ target)

Buyer viewpoint: Solar / Wind / BESS are contracted energy at fixed ■/kWh. No generation-plant MW, CAPEX or OPEX is modelled in the bill.

2. Architecture comparison (Run Analysis)

Metric	DISCOM baseline	CAPTIVE	HYBRID
Power cost ■/kWh	■9.5484/kWh	■2.7890/kWh	■4.8949/kWh
Blended energy rate	■8.7500/kWh	■2.7733/kWh	■4.6835/kWh
Annual bill	■731.89 Cr	■213.78 Cr	■375.19 Cr
Savings vs DISCOM (Y1)	—	■458.11 Cr	■311.70 Cr
NPV of savings	—	■6,717.57 Cr	■4,699.95 Cr
Contracted RE / CFE mix %	0.0%	100.0%	75.0%
Target Grid RE %	100% 0%	0% 100%	25% 75%
Hourly CFE min %	0.0%	15.1%	11.3%
Hours ≥ CFE target	0.0%	29.0%	26.7%
Hourly CFE Pass/Fail	Fail	Fail	Fail

Metric	DISCOM baseline	CAPTIVE	HYBRID
Carbon saved (tCO ₂ e)	—	628,530	471,398

DISCOM is the 100% grid baseline for savings / NPV only — not a selectable architecture. Captive and Hybrid are analysed equally from Project Setup profiles.

3. Captive — economics & mix

Item	Value
Power % (G S W B)	0.0% 33.3% 33.3% 33.3%
Target power % (Grid RE)	0.0% 100.0%
Simulated energy share (Grid RE)	0.0% 100.0%
DISCOM rate	■8.7500/kWh
Solar stack total	■4.1600/kWh
Wind stack total	■4.1600/kWh
Storage tariff (BESS)	■0.0000/kWh
Blended energy rate	■2.7733/kWh
Power cost (delivered)	■2.7890/kWh
Annual load	766,500 MWh
Annual bill	■213.78 Cr
Savings vs DISCOM (Year 1)	■458.11 Cr
NPV of savings	■6,717.57 Cr
Carbon — baseline / actual / saved	628,530 / 0 / 628,530 tCO ₂ e

Blended ■/kWh = Discom×G% + Solar×S% + Wind×W% + Storage×B% (Architecture contracted mix).

Captive — Hourly CFE & compliance

Metric	Value
Hourly CFE target	90%
Min hourly CFE %	15.1%
Mean hourly CFE %	57.7%
Hours ≥ target %	29.0%
Hourly CFE status	Fail
Annual RE % (dispatch diagnostic)	58.4%
Annual RE status	Fail

Pass = min hourly CFE ≥ target and 100% of hours meet the target. Hourly CFE% = $\text{Min}(\text{Load}, \text{Solar} + \text{Wind} + \text{CF-origin BESS}) / \text{Load} \times 100$. Economics use contracted mix × rates; profile-based CFE is reported separately.

3. Hybrid — economics & mix

Item	Value
Power % (G S W B)	25.0% 35.0% 25.0% 15.0%
Target power % (Grid RE)	25.0% 75.0%
Simulated energy share (Grid RE)	25.0% 75.0%
DISCOM rate	■8.7500/kWh
Solar stack total	■4.1600/kWh
Wind stack total	■4.1600/kWh
Storage tariff (BESS)	■0.0000/kWh
Blended energy rate	■4.6835/kWh
Power cost (delivered)	■4.8949/kWh
Annual load	766,500 MWh
Annual bill	■375.19 Cr
Savings vs DISCOM (Year 1)	■311.70 Cr
NPV of savings	■4,699.95 Cr
Carbon — baseline / actual / saved	628,530 / 157,132 / 471,398 tCO■

Blended ■/kWh = Discom×G% + Solar×S% + Wind×W% + Storage×B% (Architecture contracted mix).

Hybrid — Hourly CFE & compliance

Metric	Value
Hourly CFE target	90%
Min hourly CFE %	11.3%
Mean hourly CFE %	52.2%
Hours ≥ target %	26.7%
Hourly CFE status	Fail
Annual RE % (dispatch diagnostic)	53.1%
Annual RE status	Fail

Pass = min hourly CFE ≥ target and 100% of hours meet the target. Hourly CFE% = $\text{Min}(\text{Load}, \text{Solar} + \text{Wind} + \text{CF-origin BESS}) / \text{Load} \times 100$. Economics use contracted mix × rates; profile-based CFE is reported separately.

4. Methodology (commercial buyer)

- Bill = annual load × blended ■/kWh (+ demand / fixed / compliance if configured).
- Blended = Discom×G% + Solar×S% + Wind×W% + Storage tariff×B% from Architecture mix.
- Savings vs DISCOM = 100% DISCOM bill – architecture bill (Year 1).
- NPV of savings = present value of yearly bill savings over project life at the financial discount rate (no plant CAPEX at Year 0).
- BESS on Captive/Hybrid = Storage tariff ■/kWh only (no plant MW/MWh).

5. Disclaimer

This model is a decision-support tool. Regulatory applicability, legal interpretation, commercial eligibility and final compliance requirements should be independently validated by the appropriate legal, regulatory and commercial advisors.

Assumptions register and full methodology are available in the application. This PDF is a point-in-time export of the latest Run Analysis results.